

1)  $w$  contains at least three 1s

$$S \rightarrow 0S / 1A$$

$$A \rightarrow 0A / 1B$$

$$B \rightarrow 0B / 1C$$

$$C \rightarrow 0C / 1C / \Lambda$$

2)  $|w|$  is odd and '0' in the middle

~~$$S \rightarrow aSb / aSa / bSa / bSb /$$~~

$$S \rightarrow 0S1 / 0S0 / 1S1 / 1S0 / 0$$

$$4) 0^i 1^j ; i \leq j$$

$$S \rightarrow 0S1 / S1 / \Lambda$$

$$5) 0^i 1^j ; i < 2j$$

$$S \rightarrow S1 / 00S1 / 01 / 0S1$$

$$6) 0^i 1^j ; j = 2i$$

$$S \rightarrow 0S11 / \Lambda$$

$$7) a^i b^j c^k ; i, j, k \geq 0 ; i = j \vee j = k$$

$$S \rightarrow \cancel{aSb} A / B / \Lambda$$

$$A \rightarrow \cancel{aAb} / C$$

$$C \rightarrow \cdot$$

$$S \rightarrow A / B / \Lambda$$

$$A \rightarrow X_{AB} C$$

$$X_{AB} \rightarrow a X_{AB} b / \Lambda$$

$$C \rightarrow Cc / \Lambda$$

$$B \rightarrow X_A X_{BC}$$

$$X_{BC} \rightarrow b X_{BC} c / \Lambda$$

$$X_A \rightarrow a X_A / \Lambda$$

$$8) a^i b^j c^k ; i+j=k$$

$$S \rightarrow X_{AB} S c / \Lambda$$

$$X_{AB} \rightarrow a/b$$

$$S \rightarrow a S c / B / \Lambda$$

$$B \rightarrow b B c / \Lambda$$

$$9) a^i b^j c^k ; i+k=j$$

$$S \rightarrow AB$$

$$A \rightarrow a A b / \Lambda$$

$$B \rightarrow b B c / \Lambda$$

$$10) a^i b^j ; i=3j+2$$

$$S \rightarrow a a a S b / \text{baa}$$

| j | i  |
|---|----|
| 0 | 2  |
| 1 | 5  |
| 2 | 8  |
| 3 | 11 |

$$11) a^{2i} b^{3j} ; i, j \geq 0$$

$$S \rightarrow A / B / \Lambda$$

$$A \rightarrow a a A / \Lambda$$

$$B \rightarrow B b b b / \Lambda$$



12)  $|w|$  is odd

$$\begin{aligned} S &\rightarrow aA / bA \\ A &\rightarrow aS / bS / \Lambda \end{aligned}$$

13)  $|w| \% 3 = 0$

$$\begin{aligned} S &\rightarrow aA / bA / \Lambda \\ A &\rightarrow aB / bB \\ B &\rightarrow aS / bS \end{aligned}$$

14)  $(ab)^m c^n d^n (c)^m \quad m, n \geq 0$

$$\begin{aligned} S &\rightarrow abBc / \Lambda \\ A &\rightarrow cAd / cd \\ B &\rightarrow abBC / A \end{aligned}$$

15)  $(a^i b^i)^m (b^j a^j)^m$

$$\begin{aligned} S &\rightarrow ASB / \Lambda \\ A &\rightarrow aAb / \Lambda \\ B &\rightarrow bBa / \Lambda \end{aligned}$$

16)  $a^m b^n c^m \quad m, n \geq 2$

$$\begin{aligned} S &\rightarrow AB \\ A &\rightarrow aAb / aa bb \\ B &\rightarrow Bc / cc \end{aligned}$$

$$17) a^n b^m c^n$$

$$S \rightarrow a S c / a b b c c / \Lambda$$

$$B \rightarrow b B / \Lambda$$

$$18) a^n b^m \quad n+m = \text{even}$$

$$S \rightarrow a A / b A / \Lambda$$

$$A \rightarrow b S / a S$$

$$B \rightarrow b S / a S$$

$$19) a^n b^m \quad n+m = \text{odd}$$

$$S \rightarrow a A / b A$$

$$A \rightarrow b S / a S / \Lambda$$

$$20) \text{ Non Palindromic}$$

$$S \rightarrow a S a / b S b / a A b / b A a$$

$$A \rightarrow a A / b A / \Lambda$$

Twice as many a's as b's

$S \rightarrow aSaaSb \mid aSbSaS \mid \epsilon$

more zeroes than 1's:

$S \Rightarrow S0S0S1S \mid \Lambda \mid S1S0S0S \mid S0S1S0S$

$S \rightarrow S0S0S1S \mid S1S0S0S \mid S0S1S0S \mid S1S0S0S \mid S0S1S0S \mid \Lambda$